- Horizontal Position knob: Controls horizontal position of waveform on the screen.

Trigger controls

The trigger control delays the time base circuit operation. This makes the moving wave appears to be static (unmoved) on the screen. The frequency, wave length and amplitude can be measured, when it is unmoved (locked).

10.3.1. Working of CRO

Figure 10.21 shows block diagram of a CRO. As like TV Receivers, a Vacuum cathode Ray tube is employed as a screen in a C.R.O. The cathode of the CRT is heated using the filament. Due to thermionic emission electron ray emitted from cathode. This electron ray is controlled by the first grid/anode. The second grid accelerates the speed of the electron and the third grid focusing the electron ray at the centre of the screen. Next to the third grid, there are two pair of plates named as X and Y. These plates are termed as horizontal and vertical deflection plates. By giving a voltage to these plates the electron ray can be deflected. This is known as Electrostatic Deflection.

The X plates are connected with time base circuits. This circuit generates saw tooth wave. This saw tooth wave makes to move the (electron ray) bright spot from left to right on the screen. This is termed as X axis on the screen. The amplifier connected with Y plates, moves the ray up and down. This is termed as Y axis on the screen. The signal wave to be measured is given at the input. The switch ac/dc must be in off state. Through the capacitor the wave is amplified by the Y amplifier and fed to the Y plates.

The power of the Y amplifier can be controlled through the front panel control Volts/Div. Due to this, the wave form appears on the screen can be maximized

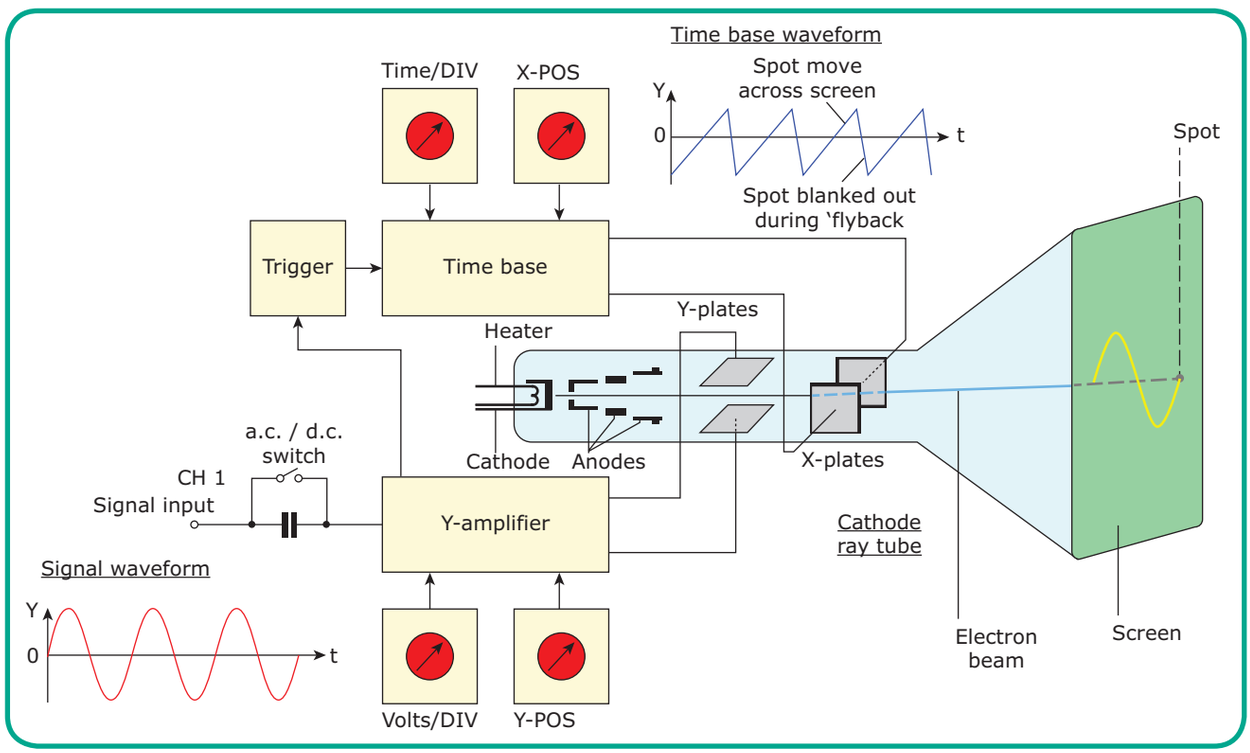


Figure 10.21 Block Diagram of a CRO

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